PROBLEM STATEMENT

Shared Resource Reservation in Medical Laboratories

Resource Reservation Management System · PrimeWave Lab (Pvt) Ltd, Software Engineering Internship technical assessment

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1 Problem statement

> Medical laboratories share a limited set of instruments, biosafety cabinets, rooms and, in lab networks, sample-transport vehicles among many staff and sections. When their use is coordinated through paper logbooks, sign-up sheets, phone calls and chat groups, nobody has one up-to-date view of what is free, what is booked and what is out of service. As a result, bookings can clash, work can be planned on equipment that is broken or due for maintenance, staff may not be told when a booked resource becomes unavailable, and managers cannot easily see which resources are overloaded and which sit idle. This costs staff time, can delay test results and makes complete equipment records hard to keep, at a time when labs face rising workloads, staff shortages and tight budgets [4, 8, 9].

Our question: How can we give every lab staff member one trustworthy, live view of shared resources, so that bookings never clash, unusable equipment is never booked, and managers can see how resources are really used?

Our answer: a web-based Resource Reservation Management System that gives all lab staff one live schedule, makes clashing bookings impossible, keeps broken and under-maintenance equipment off the schedule, and shows managers how each resource is used (Section 8).

A morning without a shared schedule *(illustrative example, not a documented incident)*

08:30	A Molecular Diagnostics technologist writes “PCR, 10:00–12:00” in the paper logbook beside the real-time PCR machine.
09:15	A technologist at a branch lab phones the main lab, hears that the machine “looks free after 11”, and sends samples for an 11:00 run.
10:30	The service engineer arrives for a calibration visit that was agreed by phone with the lab manager.
Result	Two runs and a service visit compete for one machine. One run waits until the afternoon, and no record shows how often this happens.

2 Background

Medical laboratories test patient samples so that doctors can diagnose and treat. They work in hospitals, private diagnostic networks, and public health and reference services. A typical lab has sections such as Clinical Biochemistry, Haematology, Microbiology, Molecular Diagnostics and Histopathology. Many labs work as networks, where collection centres and branch labs send samples to a main lab.

The main analysers run all day for routine tests, so they are usually not booked. Many other resources, however, are shared, and staff take turns to use them:

- **Instruments for runs or special tests:** real-time PCR machines, nucleic-acid extraction systems, HPLC analysers (for example, for HbA1c), ELISA readers and washers, flow cytometers and fluorescence microscopes.
- **Safety and storage spaces:** biosafety cabinets and cold rooms.

- **Rooms:** training and meeting rooms.
- **Vehicles:** vans or motorbikes that carry samples between collection centres, branches and the main lab.

These resources are limited and expensive, so labs must manage them carefully. WHO's laboratory quality handbook tells labs to "designate those authorized to use the equipment and when it is to be used", to schedule routine service "so as not to interrupt the flow of work", and to keep a logbook for each instrument [1]. ISO 15189, the international standard for medical laboratories, requires equipment to be operated by trained, authorized and competent staff, faulty equipment to be taken out of service and clearly labelled, and records to be kept for each item of equipment [2].

3 Why this problem is common to medical labs everywhere

- **Same rules everywhere.** WHO guidance and ISO 15189 are international, so their equipment expectations are the same wherever they are used [1, 2]. In 2024, more than 114,600 laboratories worldwide (testing, calibration and medical labs together) were accredited by bodies in the ILAC international recognition arrangement [3].
- **Same kinds of shared resources.** Hospital labs, private networks, and public health and reference labs typically share instruments, cabinets and rooms, and many move samples between sites. In Sri Lanka, a cluster laboratory system already runs in the Kalmunai health region [9], and primary care institutions send samples to apex laboratories [8]. Sample transport delays are also reported in Nepal and the Philippines [4].
- **Same pressures.** Labs in Sri Lanka and across the Asia–Pacific report rising demand, staff shortages and limited funding and equipment [4, 8, 9]. Lean studies from the USA and Spain also found manual steps, such as data processing and sample registration, to be waste [5], so this is not only a problem for low-resource labs.
- **One design fits many labs.** Each lab sets its own sites, sections, resource categories, booking limits and time zone, so the same system can serve a single hospital lab or a national lab network.

4 Who is affected

Who	What they need	What goes wrong without a shared schedule
Medical laboratory technologists and scientists	To see quickly whether a resource is free and working, and to reserve it	They check logbooks, walk to the instrument or phone around. Clashes are found on arrival, and runs wait.
Section heads and supervisors	To plan each day's runs across shared resources	Time goes on settling clashes and re-planning work.
Lab manager and quality manager	To use costly equipment well, keep equipment records for audits and plan purchases	No reliable usage data. Records are spread across logbooks and messages.
Equipment officer or biomedical engineer	To schedule service without disrupting work, and to take faulty equipment out of service	Cannot see bookings when planning service, and has no easy way to warn everyone affected.
Doctors and patients (indirectly)	Test results on time	When work is re-planned, results come later.

5 The main problems and the evidence

Problem	What the evidence shows
P1. No single, up-to-date view of resources	Sri Lanka's Medical Research Institute (MRI) reported that its manual, paper-based system caused "incorrect data entries, data duplication, increased clerical work" and difficulty tracking samples [7]. A 2024 systematic review notes that "clerical work and phone calls do not add value to the laboratory result", which is why removing them cut waiting times in one study [5]. Digital tools are still limited: only a few Sri Lankan public institutions use a laboratory information system [9], and smaller Sri Lankan facilities may lack internet access entirely [4].
P2. Bookings clash	Paper logbooks, phone calls and chat messages cannot check a new booking against existing ones, so two people can plan the same slot without knowing it. No study we found measures how often this happens, so it must be confirmed with users (Section 10).
P3. Broken or under-maintenance equipment still gets planned, and breakdowns are not passed on	WHO tells labs to schedule routine service "so as not to interrupt the flow of work", to record when equipment was removed from service and returned to use, and to place a note on faulty equipment so all staff know it is not in use [1]. ISO 15189 requires faulty equipment to be taken out of service and clearly labelled until it is shown to work [2]. At Teaching Hospital Kalutara, problems including equipment functionality contributed to "long patient waiting times, missed investigations, and inconsistencies in report quality" [10]. In Sri Lankan primary care institutions, health workers named poor maintenance of the auto analyser among the reasons tests were unavailable [8].
P4. Nothing checks who may use an instrument	WHO says "only personnel who have been trained specifically to properly use the equipment should be authorized as operators" [1], and ISO 15189 requires equipment to be operated by trained, authorized and competent staff [2]. A name written in a logbook is usually not checked against any list of authorised users.
P5. Managers cannot see how resources are used	A 2023 overview of Sri Lankan public laboratory services notes that many institutions, including MRI, "are not using the costly machinery at night", that "the efficiency of usage in the day is also below the expected level", and that improper use increases downtime and repair costs [9]. MRI reported that its manual system made it hard to "generate timely statistical data" [7]. Lack of equipment and funding are among the challenges reported by Asia–Pacific labs [4], so the equipment a lab already has must be used well.

6 Why it matters

- **Staff time.** Clerical work and phone calls "do not add value to the laboratory result" [5]. Staff are short: 20% of Asia–Pacific survey respondents cited staff shortages [4], and medical laboratory technicians (MLTs) were present in only 3 of the 9 Sri Lankan primary care institutions studied [8]. Every hour counts.
- **Turnaround time and patients.** In one hospital, the tasks in the inpatient blood-test process added up to a theoretical minimum of 47 minutes, yet delays between tasks, caused by work schedules and policies such as batching, meant the lead time could be over 40 hours [6]. In a 2024 review of seven lean studies, changes such as barcoding and process redesign cut the studies' combined turnaround time by more than 70%, although results varied widely [5]. At Teaching Hospital Kalutara, process and equipment problems contributed to long waiting times and missed investigations [10].
- **Money.** In Sri Lanka's public sector, costly machines often stand unused at night, and improper use raises downtime and repair costs [9].
- **Quality and audits.** WHO guidance and ISO 15189 expect clear equipment status, authorised use and complete records [1, 2]. Records spread across logbooks, sheets and messages are hard to keep complete and to check.

7 Root causes

1. **Information lives in many places:** logbooks, sign-up sheets, whiteboards, phones and chat groups.
2. **Nothing checks a booking when it is made:** not for overlaps, length limits, equipment status or authorisation.
3. **Equipment status and maintenance are recorded apart from the schedule,** so the schedule does not "know" when a machine is down.
4. **Use is not recorded in a form that can be counted,** so there is no usage data.

8 What the system will do

A web-based Resource Reservation Management System that gives all lab staff one live, trustworthy schedule for shared resources. Each feature answers one or more of the problems in Section 5.

Feature	What it does	Solves
1. Resource register	Lists every shared resource with its category, section, site, location and status: active, out of service or retired. Retired resources keep their history.	P1, P5
2. Live availability	Day and week views for each resource and each site, showing bookings and blocked time.	P1
3. Clash-free booking	The database itself rejects overlapping bookings, even when two people book the same slot at the same moment. Simple rules: no bookings in the past, a maximum length per category, only working resources, and room capacity.	P2
4. Maintenance and breakdowns	Managers block time for service or mark a resource out of service. One action cancels the affected bookings and records the reason, which each person sees in "My bookings".	P3
5. Usage dashboard	Shows busy and idle resources, peak hours and use by section, so managers can plan.	P5
Optional: authorised users	Only staff authorised for a resource can book it.	P4

Roles. Staff make and manage their own bookings. Lab managers manage resources, maintenance and all bookings.

Out of scope for this version. Patient data, test orders and results (the job of a laboratory information system, or LIMS); connections to analysers; reagent and stock management; staff rosters; approval workflows; email or SMS alerts; billing.

Relationship to a LIMS. A LIMS focuses on patients, samples and results. A Sri Lankan study of LIMS needs in four private Colombo hospitals listed requirements from patient registration to report issue, plus support services such as stock management, quality control and statistics; booking shared resources was not among them [11]. This system fills that gap and could connect to a LIMS later.

9 How we will know it works

Targets for the first version, checked by automated tests and in the demo:

- Zero overlapping confirmed bookings for any resource, guaranteed by the database and proven by an automated test that sends two booking requests at the same moment.
- Zero bookings accepted for out-of-service resources or during maintenance blocks.
- Every booking cancelled by a breakdown shows its reason to the person who made it.
- A staff member can find a free slot and book it in under one minute (checked in a short user test).
- A manager can see four weeks of use by resource and by section.

10 Assumptions, limits and validation

Assumptions

- An internal tool for lab staff only. It stores no patient data.
- One organisation with several sites in one time zone. The demo uses Sri Lanka time; the time zone is a setting.
- Each physical unit is its own resource: two PCR machines are two resources.
- Main analysers that run all day are not booked.
- It runs in a web browser and needs a network connection to its server. A site without internet [4] could use an installation on its local network.

Limits of the evidence

The sources show the pressures labs face, what the standards require and the cost of manual processes. None of them measures how often booking clashes happen. The Asia–Pacific survey had a response rate of only 1.2% [4], and the Sri Lankan studies cover particular settings. So, before a real rollout, the problem should be confirmed with users: interview a lab manager, two or three technologists and an equipment officer, and ask:

- How are shared instruments, cabinets, rooms and vehicles booked today?
- How do you find out whether something is free, in use or out of service?
- How often do two people plan the same resource, and what happens then?
- How are service visits and breakdowns announced?
- What would managers like to know about how resources are used?

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